

Suggested Solutions to Exercises 2

1. The difference between a long forward and a long call option with the same strike price of £50 is that in the first case (forward) *we have to buy the asset* for £50 at the agreed time, while in the second case (option) *we have the right to buy the asset* for £50 at the specified time, *but no obligation to do that*.
2. We have £5,800 to invest and we want to speculate on a rise in the price of a certain stock (current price £29). Three-month calls with strike of £30 are selling for £2.90. We can adopt two strategies:
 - A) Buy 200 shares ($29 \times 200 = 5,800$)
 - B) Buy 2,000 options ($2.90 \times 2,000 = 5,800$)

If the share price goes up, the second strategy gives the larger gain. For example, if the price in three months is £40 then:

– Profit from A):

$$200 \times (40 - 29) = 2,200$$

– Profit from B):

$$2,000 \times (40 - 30) - 5,800 = 22,000 - 5,800 = 14,200$$

However, if the price goes down, the second strategy gives greater losses. E.g. if the price hits £25 then:

– Loss from A): $200 \times (29 - 25) = 800$

– Loss from B): 5,800 (we do not exercise the option, thus we lose everything we spent to buy the options)

Notice that for any stock price (at three months) less than £30, strategy B) will always give a constant loss of £5,800, while strategy A) gives a loss that is linear in the stock price, and equal to £5,800 (its maximum) only in the case of the stock price dropping to zero!

3. The first thing to notice is that – by definition – European options can only be exercised at maturity, and this gives a first similarity with a forward contract, i.e. action can be taken only at maturity (this would not be the case for American options).

Once verified that all actions in the three contracts will take place at the same time, we can look at the payoff diagrams, where it is evident that the payoff from a long forward contract (as a function of the price of the stock at maturity) equals the sum of the payoffs of a long call and a short put.

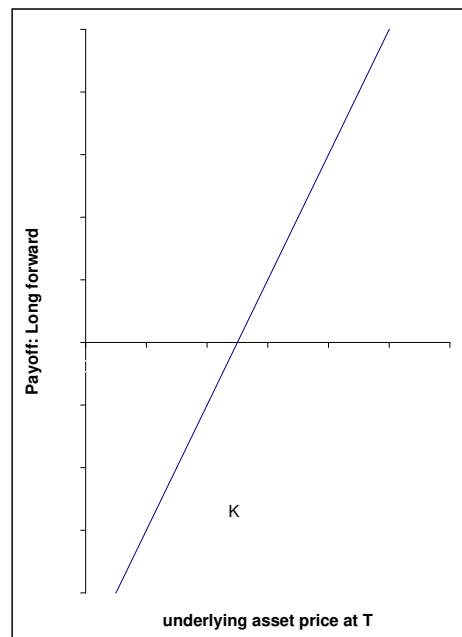


Figure 1: Payoff for long forward position, delivery price K .

We can conclude that holding a long position in a forward contract is equivalent to holding a portfolio composed by one long call option and one short put option with the same strike price (and equal to the delivery price of the forward contract).

4. The price of a forward contract on a security that provides no income is given by the future value of the underlying security, computed using

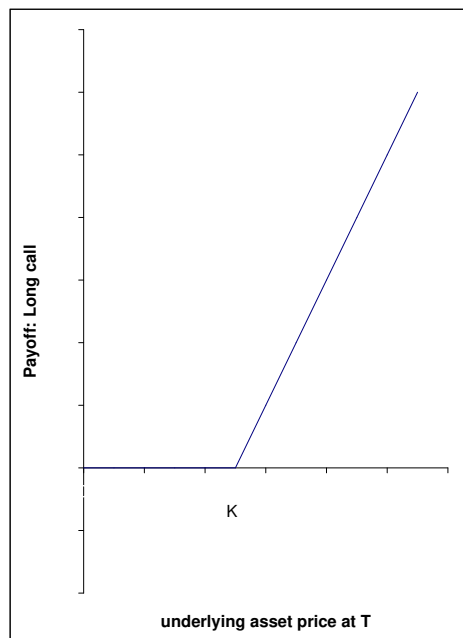


Figure 2: Payoff for long European call position, strike price K .

the risk-free interest rate:

$$F = Se^{rT}$$

In our case $S = £30$, $r = 12\%$ and the time to maturity is six months. Thus, the forward price is given by:

$$F = 30 e^{0.12 \times \frac{6}{12}} = 31.86$$

5. We have again a case of a forward contract on a security that provides no income, where the time to maturity is now one year, the risk-free rate is 10%, and the security is currently priced £40.
 - (a) The forward price is computed using the formula in Exercise No.4 above, and equals:

$$F = 40 e^{0.1 \times 1} = 44.21$$

This value will be the delivery price of the contract, and the initial value of the contract is zero by construction.

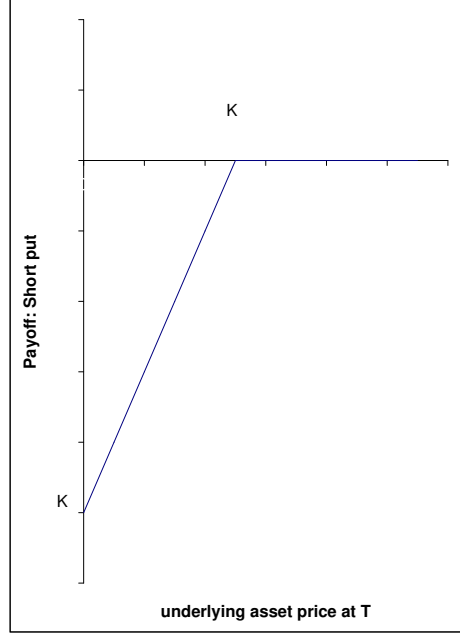


Figure 3: Payoffs short European put position, strike price K .

- (b) After six months the stock price goes up to £45, and the time to maturity is reduced to six months; the rate remains the same, i.e. 10%. The new forward price is now:

$$F = 45 e^{0.1 \times \frac{6}{12}} = 47.31$$

- (c) The value of the contract now will be equal to the value of a six-month forward contract written now with delivery price of £44.21, i.e.,

$$f = 45 - 44.21 e^{-0.1 \times \frac{6}{12}} = 2.95$$

Alternatively, the value can be computed as the difference between the forward price and the delivery price (i.e. the payoff we would get if the stock at maturity had price equal to the forward price), discounted to present. This gives:

$$f = [47.31 - 44.21] e^{-0.1 \times \frac{6}{12}} = 2.95$$

6. In this case we have a six-month forward contract on a security that provides known income (\$1) after two and five months. Since the risk-free rate $r = 8\%$ is constant for all maturities, I can use the same r to discount all the payments from the security. The present value I of the income from the security is given by:

$$I = 1 \cdot e^{-0.08 \times \frac{2}{12}} + 1 \cdot e^{-0.08 \times \frac{5}{12}} = 1.9540$$

Then the forward price can be computed using the formula:

$$F = (S - I)e^{rT}$$

which in our case gives:

$$F = (50 - 1.9540)e^{0.08 \times \frac{6}{12}} = 50.01$$

Note that the initial value of the contract is by design equal to zero.